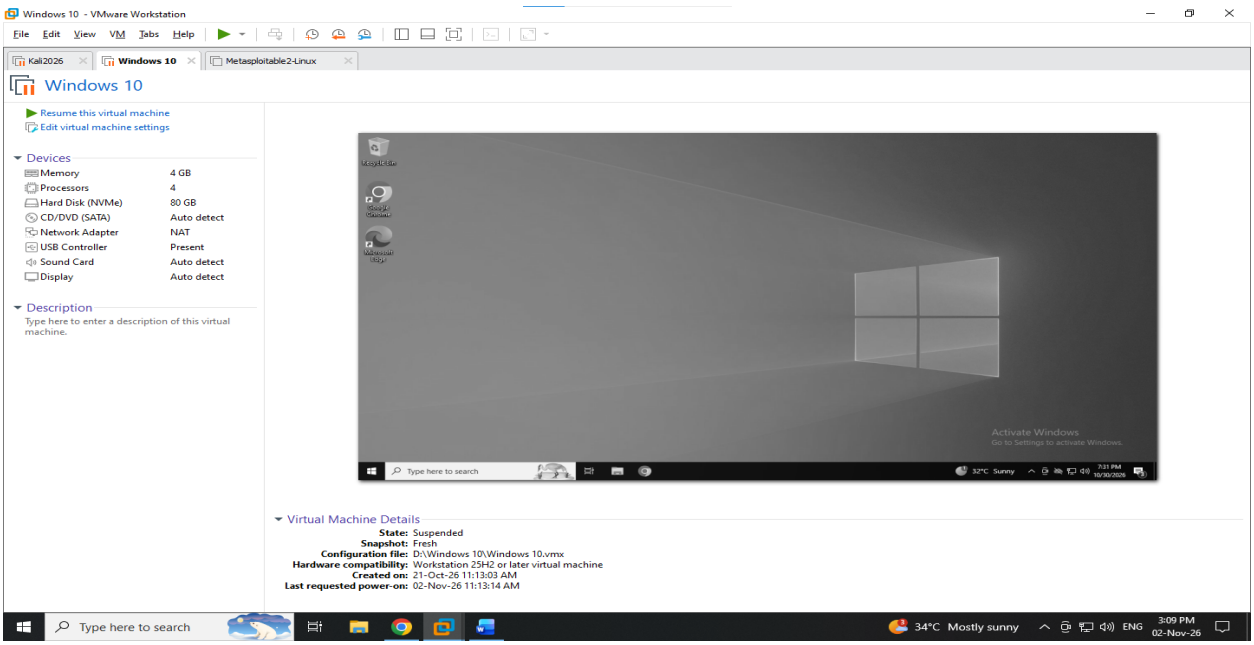
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Q.1 Install VirtualBox or VMware Workstation on your computer and take a screenshot of the installation screen showing the software is installed.

ANS :

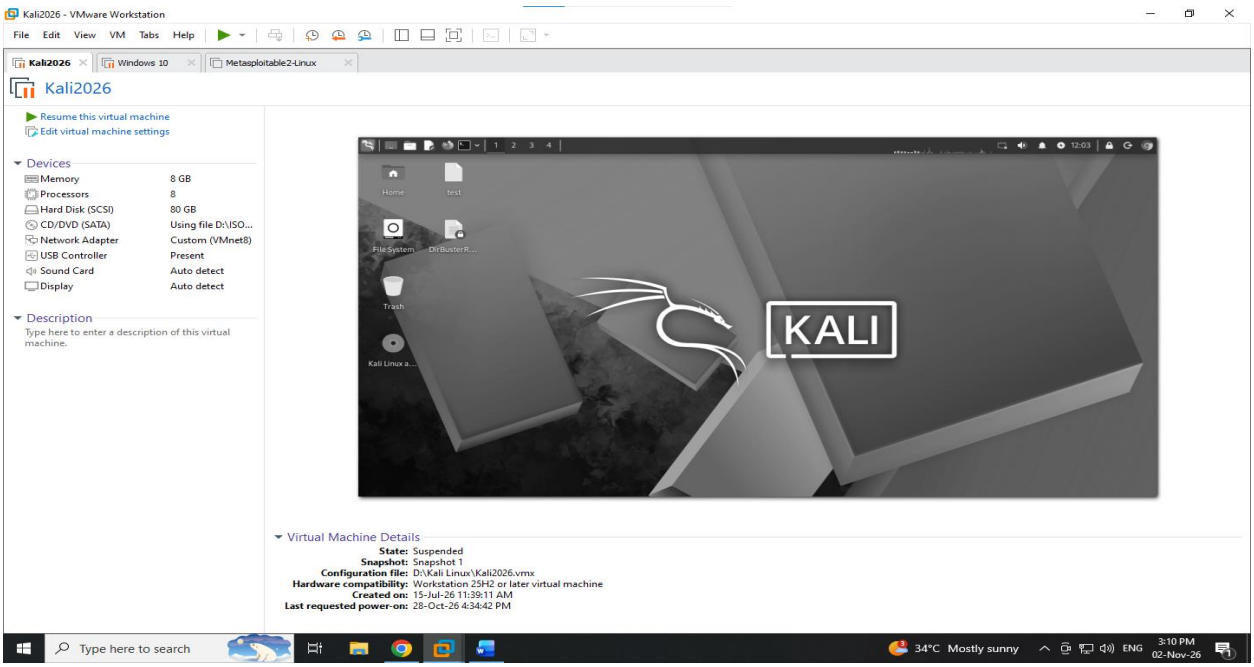


Q.2 Create a new virtual machine in VirtualBox or VMware Workstation and configure it to run Ubuntu Linux as the guest OS with at least 2GB RAM and 20GB storage. Hint: Download the Ubuntu ISO file from the official website if you don't already have it.

ANS :

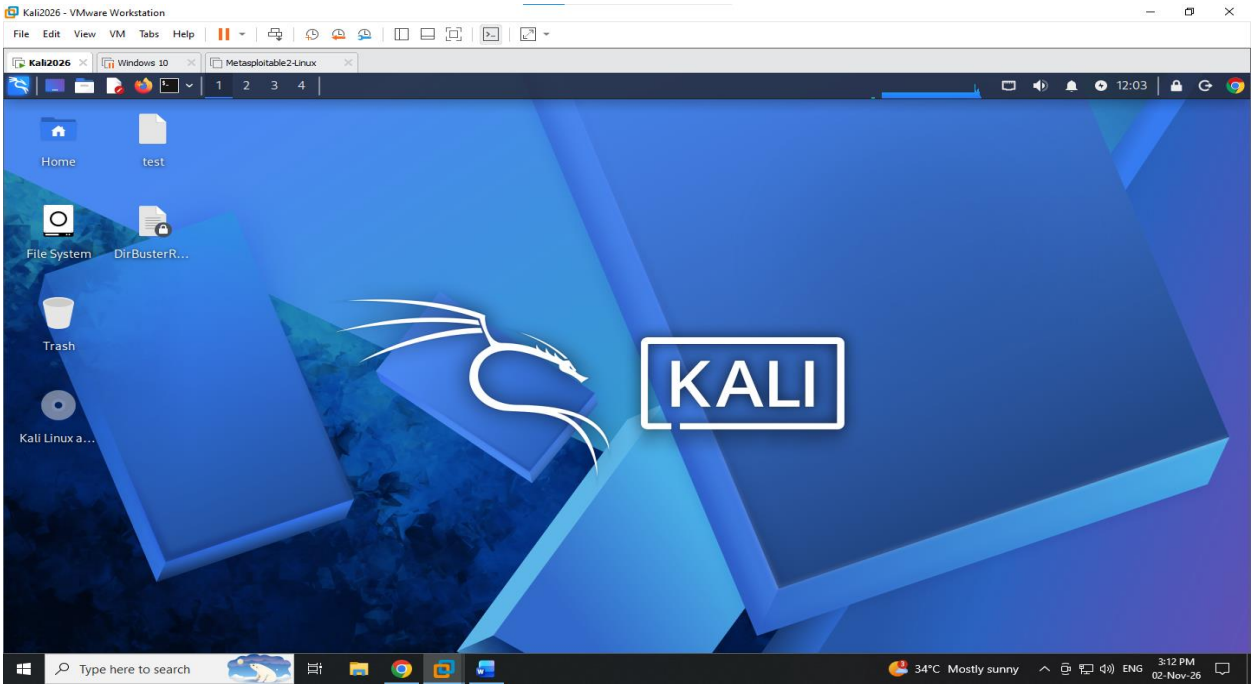
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Q.3 Start your Ubuntu virtual machine and take a screenshot of the Ubuntu desktop running inside the VM window.

ANS :



Q.4 List 3 differences between Type 1 (bare-metal) and Type 2 (hosted) hypervisors, and name one real-world example of each.

ANS : Here are three key differences between Type 1 (bare-metal) and Type 2 (hosted) hypervisors, along with a real-world example of each:

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3 Key Differences

1. **Architecture & Foundation**

    - **Type 1:** Installed directly onto the physical hardware of the host machine, acting as its own lightweight operating system.

    - **Type 2:** Installed as an application running on top of an existing host operating system (such as Windows, macOS, or Linux).

2. **Performance & Efficiency**

    - **Type 1:** Offers superior performance, low latency, and high resource efficiency because virtual machines interact directly with the hardware without intermediary overhead.

    - **Type 2:** Incurs a performance penalty and higher resource overhead because hardware requests must pass through both the hypervisor and the underlying host operating system.

3. **Primary Use Case**

    - **Type 1:** Designed primarily for enterprise data centers, server virtualization, cloud computing, and production environments requiring high availability.

- **Type 2:** Commonly used for local desktop virtualization, software testing, security research labs, and personal development environments.

Real-World Examples

- **Type 1 Example: VMware ESXi** (widely used in enterprise data centers for managing enterprise-grade server infrastructure).

- **Type 2 Example: Oracle VM VirtualBox** (popular for running virtual machines locally on desktop computers for testing and learning).